

# Exponential Expressions

---

## MA.7.NSO.1.1

*I can apply more than one law of exponents to evaluate an expression and to write an equivalent one.*

*Yesterday: seven laws, one at a time.*

**Today: two or three laws in one expression — and the order is yours.**

# Warm-Up

---

*On your own. Four minutes. No notes.*

1. Find the value:  $\frac{10^5}{10^3}$
2. Is  $\sqrt{50}$  closer to 7 or to 8?
3. Solve  $4(x - 2) = 12$
4. Evaluate  $2 + 3 \cdot 4^2$

# Warm-Up

---

*Answers.*

1. Find the value:  $\frac{10^5}{10^3}$  **100**
2. Is  $\sqrt{50}$  closer to 7 or to 8? **7**
3. Solve  $4(x - 2) = 12$   **$x = 5$**
4. Evaluate  $2 + 3 \cdot 4^2$  **50**

# Notes

*Copy the table. It is your reference for the rest of the unit.*

## I. The seven laws, in one place

| Law                 | Rule                          | Example                             |
|---------------------|-------------------------------|-------------------------------------|
| Product of powers   | $a^m \cdot a^n = a^{m+n}$     | $2^3 \cdot 2^1 = 2^4$               |
| Quotient of powers  | $a^m \div a^n = a^{m-n}$      | $(-3)^7 \div (-3)^2 = (-3)^5$       |
| Power of a product  | $(ab)^m = a^m \cdot b^m$      | $(-7 \cdot 3)^2 = (-7)^2 \cdot 3^2$ |
| Power of a quotient | $(a \div b)^m = a^m \div b^m$ | $(13 \div 5)^3 = 13^3 \div 5^3$     |
| Power of a power    | $(a^m)^n = a^{mn}$            | $(6^3)^4 = 6^{12}$                  |
| Identity exponent   | $a^1 = a$                     | $15^1 = 15$                         |
| Zero exponent       | $a^0 = 1$                     | $127^0 = 1$                         |

# Notes

---

*Two students, two orders, one value.*

## II. More than one law — make a plan, then follow it

---

$$\text{Winston: } \left( \frac{2^6}{2^4} \right)^3 = (2^2)^3 = 2^6 = 64$$

$$\text{Yvette: } \left( \frac{2^6}{2^4} \right)^3 = \frac{(2^6)^3}{(2^4)^3} = \frac{2^{18}}{2^{12}} = 2^6 = 64$$

Different plans, same value. Any correct sequence of laws gives the same answer.

# Notes

---

## III. Order of operations still applies

---

- A. Parentheses first — including a fraction bar, which is a pair of parentheses in disguise.
- B. Exponents next — the laws of exponents are applied here.
- C. Then multiply and divide, left to right; then add and subtract.
- D. **Never across a + or a -:**  $2^3 + 2^2 = 8 + 4 = 12$ , **not**  $2^5$ .

# Example 1

---

*A zero exponent inside a product inside a quotient.*

$$\frac{4^5 \cdot 4^0}{4^2}$$

**Make a plan. Then find the value.**

# Example 1

---

*Worked — Caroline's plan and Jamir's plan.*

$$\frac{4^5 \cdot 4^0}{4^2} = \frac{4^{5+0}}{4^2} = \frac{4^5}{4^2} \quad \text{Caroline: product of powers first}$$

$$\frac{4^5 \cdot 4^0}{4^2} = \frac{4^5 \cdot 1}{4^2} = \frac{4^5}{4^2} \quad \text{Jamir: zero exponent first}$$

$$\frac{4^5}{4^2} = 4^3 = 64 \quad \text{both: quotient of powers, then evaluate}$$

**Answer: 64**

# Your Turn

---

*Same steps, your numbers. Boards up when done.*

$$\frac{5^4 \cdot 5^0}{5^2}$$

# Your Turn

---

*Answer.*

$$\frac{5^4 \cdot 5^0}{5^2}$$

$5^4 \cdot 1$  over  $5^2$ , so  $5^2$

**Answer:  $5^2 = 25$**

# Example 2

---

*Three laws, a decimal and a negative base.*

$$11.6^0 \cdot \frac{(-15)^4}{1.5^4}$$

**Make a plan. Then find the value.**

# Example 2

---

*Worked.*

$$11.6^0 \cdot \frac{(-15)^4}{1.5^4} = 1 \cdot \left(\frac{-15}{1.5}\right)^4 \quad \text{zero exponent; power of a quotient, backwards}$$

$$\left(\frac{-15}{1.5}\right)^4 = (-10)^4 \quad \text{divide inside: } -15 \div 1.5 = -10$$

$$(-10)^4 = 10000 \quad \text{four negative factors: positive}$$

**Answer: 10,000**

# Your Turn

---

*Same steps, your numbers. Boards up when done.*

$$7^0 \cdot \frac{(-12)^3}{4^3}$$

# Your Turn

---

*Answer.*

$$7^0 \cdot \frac{(-12)^3}{4^3}$$

$$1 \cdot (-12 \div 4)^3 = (-3)^3$$

**Answer: -27**

# Whiteboards • Question 1 of 9

---

*Boards up on three.*

A warehouse holds  $6^4$  boxes in  $6^2$  equal rows.

Each box weighs 3 kilograms.

**How many boxes are in each row?**

**Answer it.**

# Whiteboards • Question 1 of 9

---

*Answer.*

A warehouse holds  $6^4$  boxes in  $6^2$  equal rows.

Each box weighs 3 kilograms.

**How many boxes are in each row?**

$6^4$  over  $6^2$ , so  $6^2$

**Answer: 36**

# Whiteboards • Question 2 of 9

---

*Boards up on three.*

$$(3^0 \cdot 3^2)^3$$

**Answer it.**

*Find the value.*

# Whiteboards · Question 2 of 9

---

*Answer.*

$$(3^0 \cdot 3^2)^3$$

$$(3^2)^3 = 3^6$$

**Answer: 729**

# Whiteboards • Question 3 of 9

---

*Boards up on three.*

$$\frac{(0.5^3)^2}{(0.5)^2}$$

**Answer it.**

*Find the value.*

# Whiteboards • Question 3 of 9

---

*Answer.*

$$\frac{(0.5^3)^2}{(0.5)^2}$$

$$0.5^6 \div 0.5^2 = 0.5^4$$

**Answer: 0.0625**

# Whiteboards • Question 4 of 9

---

*Boards up on three.*

$$\left(\frac{2.5^5}{2.5^3}\right)^1$$

**Answer it.**

*Find the value.*

# Whiteboards • Question 4 of 9

---

*Answer.*

$$\left( \frac{2.5^5}{2.5^3} \right)^1$$

$2.5^2$  — the exponent 1 changes nothing

**Answer: 6.25**

# Whiteboards • Question 5 of 9

---

*Boards up on three.*

$$\frac{(2^2 \cdot 3 \cdot 2^0)^6}{2^{10}}$$

Which is the value?

**A.**  $1/4$

**B.**  $2^{22}$

**C.**  $0$

**D.**  $4$

**Answer it.**

# Whiteboards • Question 5 of 9

---

*Answer.*

$$\frac{(2^2 \cdot 3 \cdot 2^0)^6}{2^{10}}$$

Which is the value?

A.  $1/4$

B.  $2^{22}$

C. 0

D. 4

**Answer: D — 4**

# Whiteboards • Question 6 of 9

---

*Boards up on three.*

$$\frac{0.2^3}{2^3} \cdot 0.1^2$$

Which expression is equivalent?

**A.**  $0.2^0 \cdot 0.1^2$

**B.**  $0.2^3 \cdot 0.1^2$

**C.**  $0.1^5$

**D.**  $0.1^6$

**Answer it.**

# Whiteboards • Question 6 of 9

---

*Answer.*

$$\frac{0.2^3}{2^3} \cdot 0.1^2$$

Which expression is equivalent?

**A.**  $0.2^0 \cdot 0.1^2$

**B.**  $0.2^3 \cdot 0.1^2$

**C.**  $0.1^5$

**D.**  $0.1^6$

**Answer: C —  $0.1^5$**

# Whiteboards • Question 7 of 9

---

*Boards up on three.*

$$0.1^2 \cdot (0.1^1)^2$$

**Answer it.**

*Find the value.*

# Whiteboards • Question 7 of 9

---

*Answer.*

$$0.1^2 \cdot (0.1^1)^2$$

$$0.1^2 \cdot 0.1^2 = 0.1^4$$

**Answer: 0.0001**

# Whiteboards • Question 8 of 9

---

*Boards up on three.*

$$\frac{(-4)^2 \cdot (-4)^1}{(-4)^0}$$

**Answer it.**

*Find the value.*

# Whiteboards • Question 8 of 9

---

*Answer.*

$$\frac{(-4)^2 \cdot (-4)^1}{(-4)^0}$$

$(-4)^3$  over 1

**Answer: -64**

# Whiteboards • Question 9 of 9

---

*Take your time. Boards up when you have written it.*

Winston writes  $\left(\frac{2^6}{2^4}\right)^3 = (2^2)^3$ . Yvette writes  $\left(\frac{2^6}{2^4}\right)^3 = \frac{2^{18}}{2^{12}}$ .

Do they get the same value? Explain.

**Write your answer in sentences.**

*Name the law each one used first, and give the value.*

# Whiteboards • Question 9 of 9

---

*Answer.*

Winston writes  $\left(\frac{2^6}{2^4}\right)^3 = (2^2)^3$ . Yvette writes  $\left(\frac{2^6}{2^4}\right)^3 = \frac{2^{18}}{2^{12}}$ .

Do they get the same value? Explain.

both plans end at  $2^6$

**Answer: Yes — both give  $2^6 = 64$**

# Independent Set

---

*Six questions. On your own, in silence.*

1. Six questions. Work down the page.
2. Show the step that does the work, not just the answer.
3. Silence until the six minutes are up.

# IXL

---

*Last five minutes.*

1. Open IXL and start today's skills.
2. Work on paper where the question needs work. The answer box does not show it.
3. Every skill listed is required, to a SmartScore of 67. Due at the start of the next class.

## **Today's skills — all required**

Evaluate expressions using properties of exponents — ZE9